Google Cloud Platform (GCP) Training Program

Kickoff & orientation

PROGRAM · OVERVIEW

What this program covers

Duration

5 days · 5 hours per day · 25 hours

Format

Concept walkthroughs, architecture diagrams, hands-on labs, and a daily quiz

Scope

Storage & ingestion through BigQuery, orchestration, Vertex AI, and MLOps

Capstone

A project to done post session GCP data + AI project after Day 5

- 25 hours total
- Primary focus on GCP
- Data + AI on GCP
- Capstone Finale

Program map

1	**GCP Foundations** Storage & Ingestion · BigQuery Basics	5 hrs
2	**BigQuery Advanced & Governance** External Tables & BQML · RLS, Lineage & Security	5 hrs
3	**Orchestration, Pipelines & Security** Composer, Dataflow, Dataproc · Data Catalog & Policy Tags	5 hrs
4	**Vertex AI Platform & Gen AI** AutoML, Feature Store, Pipelines · Gemini, RAG, Agent Builder	5 hrs
5	**MLOps & Infra Reliability** CI/CD for ML, Model Monitoring · GKE, Multi-Project Topology	5 hrs
★	**Capstone Project** End-to-end secure GCP data + AI implementation	Post Session

What is a GCP project?

Project

The basic container for resources, billing, and IAM — almost everything you create lives inside one. You enable services by enabling APIs.

Project ID

A globally unique identifier you'll reference constantly in commands and consoles

One project, many resources

Storage buckets, BigQuery datasets, Vertex AI models — all scoped to a project

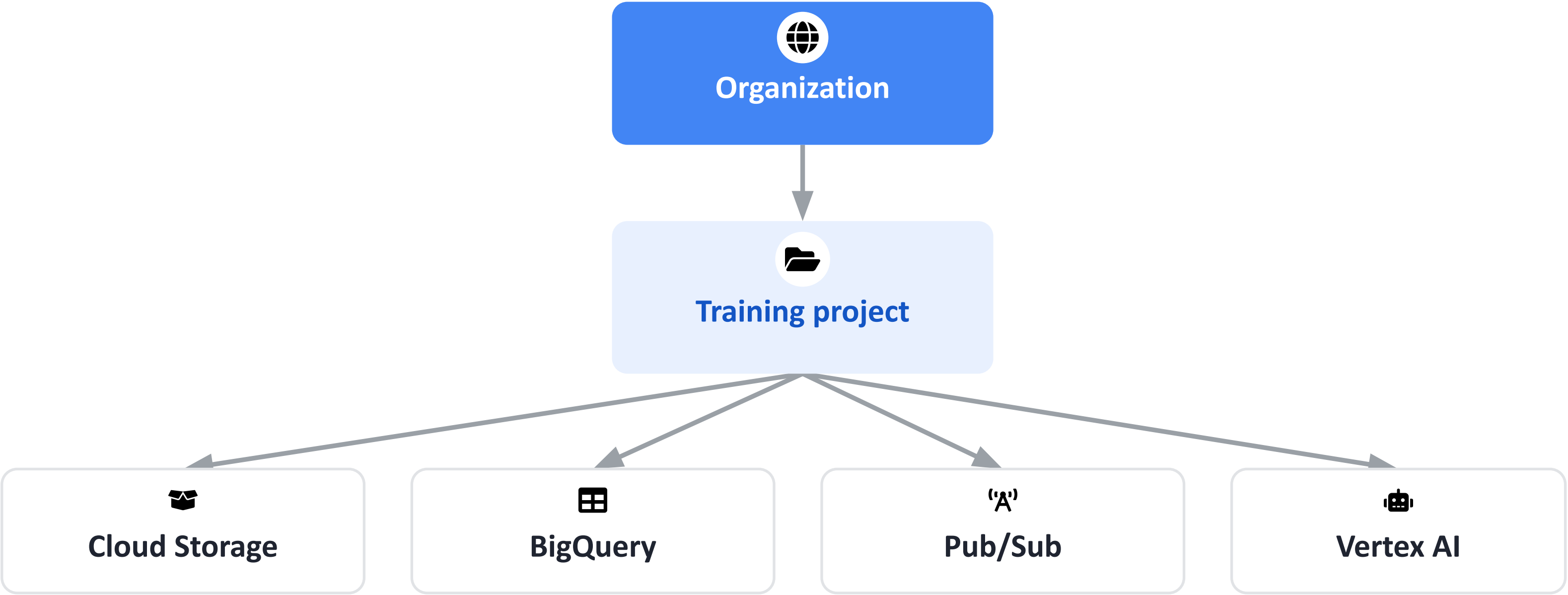
Project Boundary

Anything and everything that you do will always below to a project in GCP. Only a handful of resources can be shared across projects.

- Project
- Project ID
- Resources
- Shared project

GOOGLE CLOUD OVERVIEW

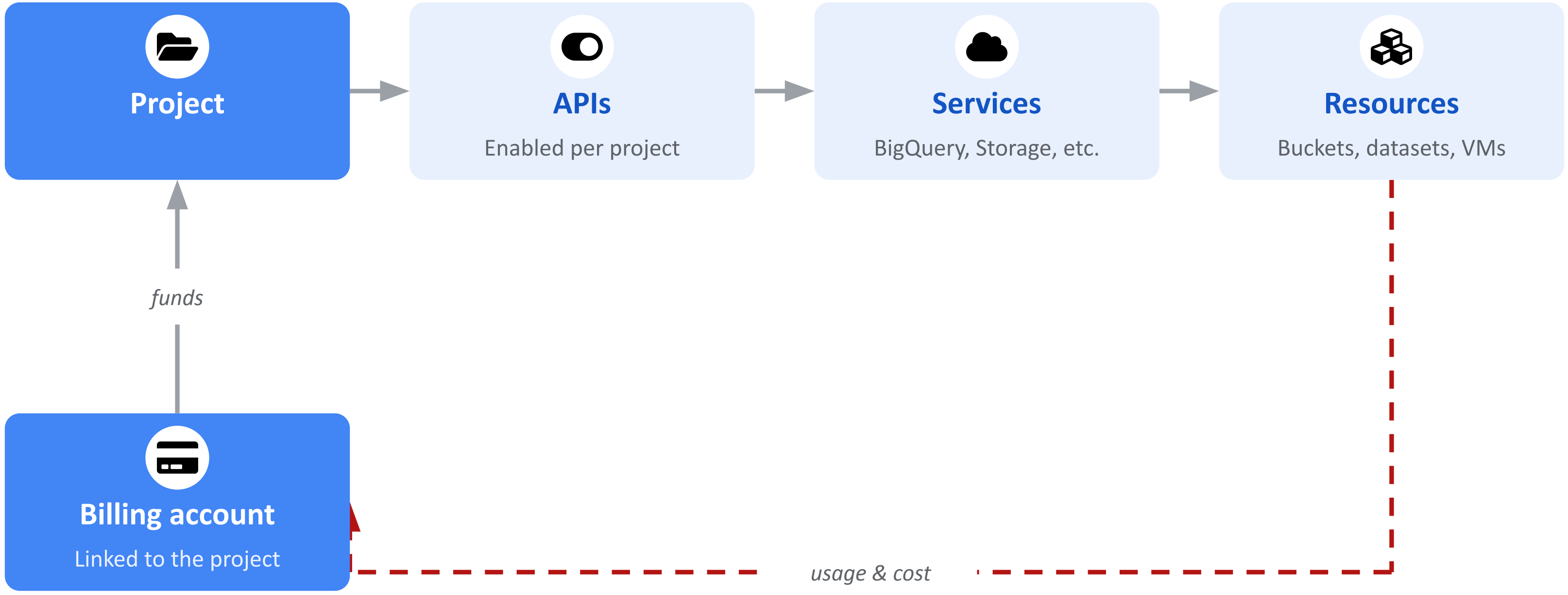
Where everything you'll use this week lives



Everything you build/do in gcp happens in a project, its upto you how you choose to structure.

GOOGLE CLOUD OVERVIEW

How it all connects: project, APIs, services, resources, billing



Billing funds the project. The project enables APIs, which unlock services, which let you create resources — and every resource's usage flows back to billing.

GOOGLE CLOUD OVERVIEW

IAM: the core access model

Principal

Who is asking — a user, a group, or a service account

Role

A named bundle of permissions, like Viewer or Editor

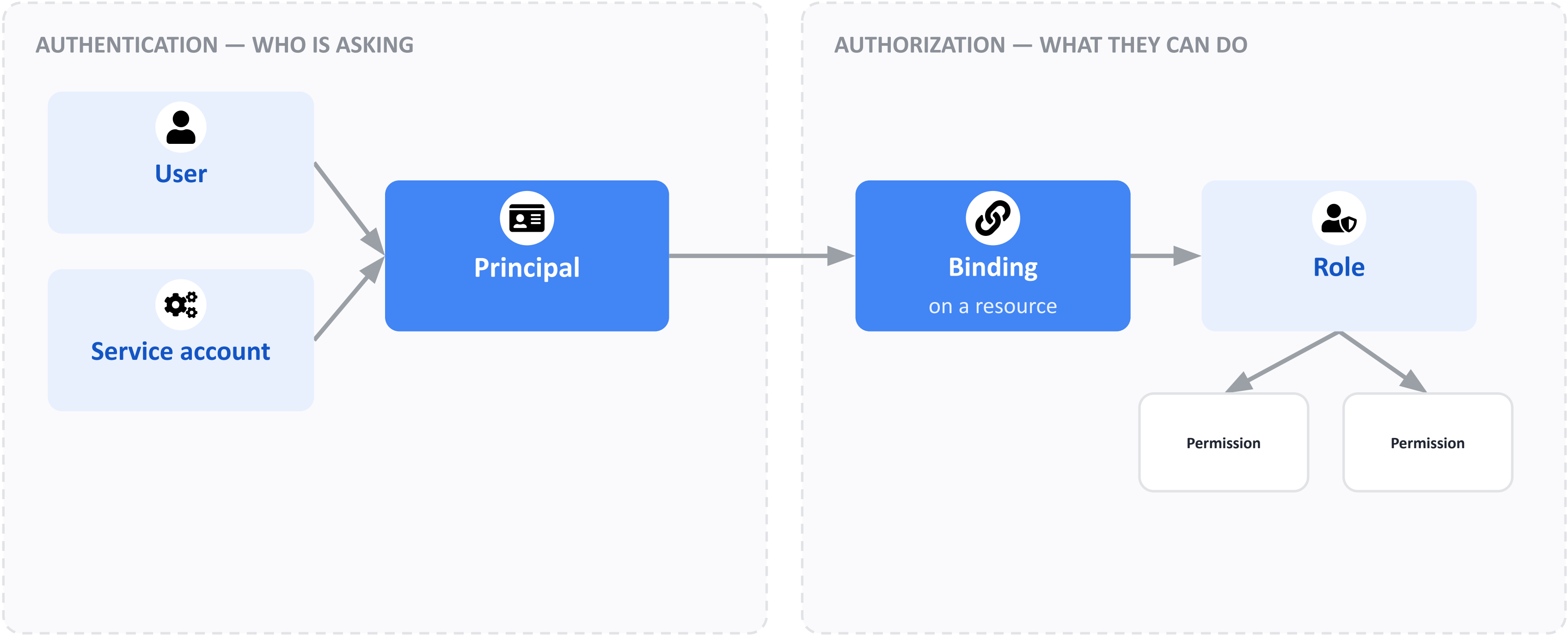
Permission

A single allowed action, like 'read a BigQuery table'

Policy binding

Ties a principal to a role on a specific resource

Principals, roles, permissions & bindings



Authentication proves who's asking (the Principal). Authorization decides what they can do — a Binding grants a Principal a Role, and a Role is just a bundle of Permissions.

IAM Roles - Authorisation

Role	Scope	What it lets you do
Viewer	Project or resource	Read-only — see resources and data, change nothing
Editor	Project or resource	Read and write — create, modify, and delete most resources
Owner	Project	Full control, including managing who else has access
Service-specific roles	One service only	Scoped access to just that service (e.g. BigQuery Data Editor)

IAM BASICS

IAM basics bigquery: who can do what

Viewer

Read-only access to data and metadata — no changes allowed

Editor

Read and write access to data within a dataset or bucket

Admin

Full control, including managing permissions for others

Job User

Can run queries and jobs without necessarily owning the data

Finding your way around the GCP Console

1. **Project switcher**
   Top left — confirms which project you're currently working in
2. **Navigation menu**
   Top left hamburger icon — every GCP service is listed here
3. **Search bar**
   Top center — fastest way to jump to any service or resource by name
4. **Cloud Shell**
   Top right terminal icon — a free, pre-authenticated command line, no install needed
5. **Notifications & billing**
   Top right — budget alerts and a quick link to current spend

Try it now: Cloud Shell & Colab against your project

1. **Open Cloud Shell**

```
Console → top-right terminal icon (>_)
```

2. **Point it at your project**

```
gcloud config set project <PROJECT_ID>
```

3. **Run something to prove it**

```
gcloud projects describe $(gcloud config get-value project)
```

4. **Open Colab**

```
colab.research.google.com → New notebook
```

5. **Authenticate as yourself**

```
from google.colab import auth; auth.authenticate_user()
```

6. **Run a query against the project**

```
!bq query --project_id=<PROJECT_ID> 'SELECT 1'
```

GOOGLE CLOUD OVERVIEW

Google Cloud SDK for Python:

Client libraries, not one SDK

A separate Python package per service — google-cloud-bigquery, google-cloud-storage, and so on

Different from gcloud

gcloud is the command-line tool; client libraries are what your Python code actually imports

Same authentication underneath

Both use the same identity — Application Default Credentials, already set up in Colab

Officially maintained

Built and supported by Google, with consistent patterns across every service

Python

Per-service packages

Same auth (ADC)

Google-maintained

How it works: from your code to the API

Your code → **Client library** → **ADC** → **REST / gRPC** → **GCP service**

- **Your code**: import google.cloud
- **Client library**: e.g. bigquery.Client()
- **ADC**: Finds credentials automatically
- **REST / gRPC**: API call over HTTPS
- **GCP service**: BigQuery, Storage, etc.

The client library handles auth, retries, and serialization — you just call a Python method.

How to use it: install, authenticate, call

1. **Install the library you need**

```
pip install google-cloud-bigquery google-cloud-storage
```

2. **Authenticate (already done in Colab)**

```
from google.colab import auth; auth.authenticate_user()
```

3. **Create a client**

```
from google.cloud import bigquery; client =
bigquery.Client(project='my-project')
```

4. **Call a method like any Python object**

```
df = client.query('SELECT * FROM ds.orders LIMIT
10').to_dataframe()
```

Documentation & resources - https://docs.cloud.google.com/python/docs/reference

Official docs

cloud.google.com/python/docs — a getting-started guide for every service

API reference

Full method-by-method reference, generated directly from the source code

Gcloud Command Line

A CLI which can be used to run commands

PyPI packages

pypi.org — search “google-cloud-” to find the package for any service

Billing basics

Billing account

The payment method tied to a project — every active project needs one linked

Budgets & alerts

Configurable thresholds that email you before spend gets out of hand

Costing

Cost of a resource will depend upon location, scale and the type of resources you create at large, though certain services can have other billing models.

Why it matters

Budgets can overshoot quickly

- Billing account
- Budget alerts
- Sandbox covered
- Track spend

GETTING STARTED · LAB ACCESS

How to access your GCP lab environment

1. Go your gmail or project invite
2. Sign in at console.cloud.google.com
3. Verify project & billing access
4. Set up Google Colab
5. Install the gcloud CLI or Access the cloud shell, projects, services, resources

HOW WE'LL RUN EACH DAY

Session methodology

Concept walkthroughs

Short, focused explanations of each service before going hands-on

Architecture diagrams

Every hard concept gets a real visual — not just bullet points

Hands-on labs

Real gcloud, bq, and SQL commands you run yourself, not just watch

Daily recap quiz

A live Kahoot quiz closes every day to check what actually stuck

RESOURCES

Where to find everything

Daily slide decks

One deck per day, shared after each session — yours to keep

Lab command reference

Every hands-on command lives in that lab slide's notebook

Google Colab

colab.research.google.com — this week's notebook environment

Notebooks

Github Repository contains all

DAILY QUIZ

How the daily Kahoot quiz works

When

Last 10 minutes of every day, right after the closing recap

How to join

Go to kahoot.it on your phone or laptop and enter the game PIN shown on screen

Format

10 questions, 10 minutes, covering that day's concepts

- Join on your phone
- 10 min, 10 questions
- Live leaderboard
- Checkpoint, not a test

Before we start today's quiz...

Download the Kahoot! app

Scan to open kahoot.it

- Search “Kahoot!” on the App Store
- Search “Kahoot!” on Google Play
- No app? Just open kahoot.it in any browser
- When we start, enter the Game PIN shown on screen

Let's get to know each other

A quick round of introductions before we dive into Day 1

INTRODUCTION

Your name

What you do

Your role, team, or what you're hoping to use GCP for

One question

Anything on the agenda or sessions you're curious about

Ready for Day 1

GCP Foundations — Storage, Ingestion & BigQuery Basics starts now

Questions?